



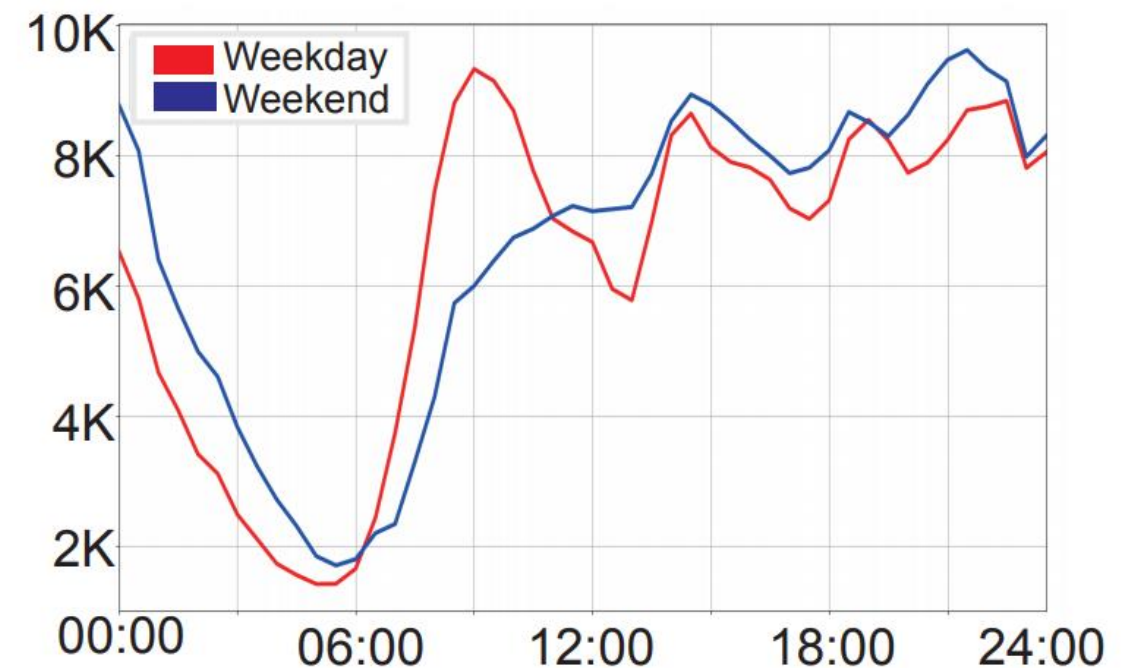
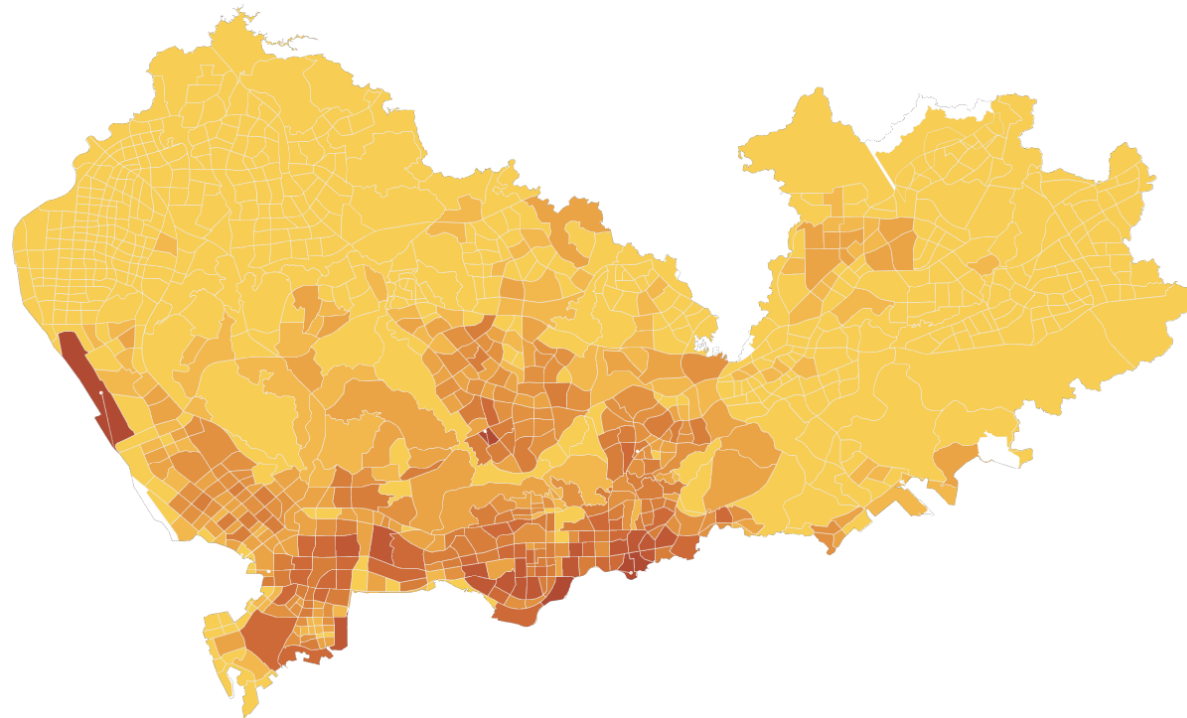
Revisiting the Modifiable Areal Unit Problem in Deep Traffic Prediction with Visual Analytics

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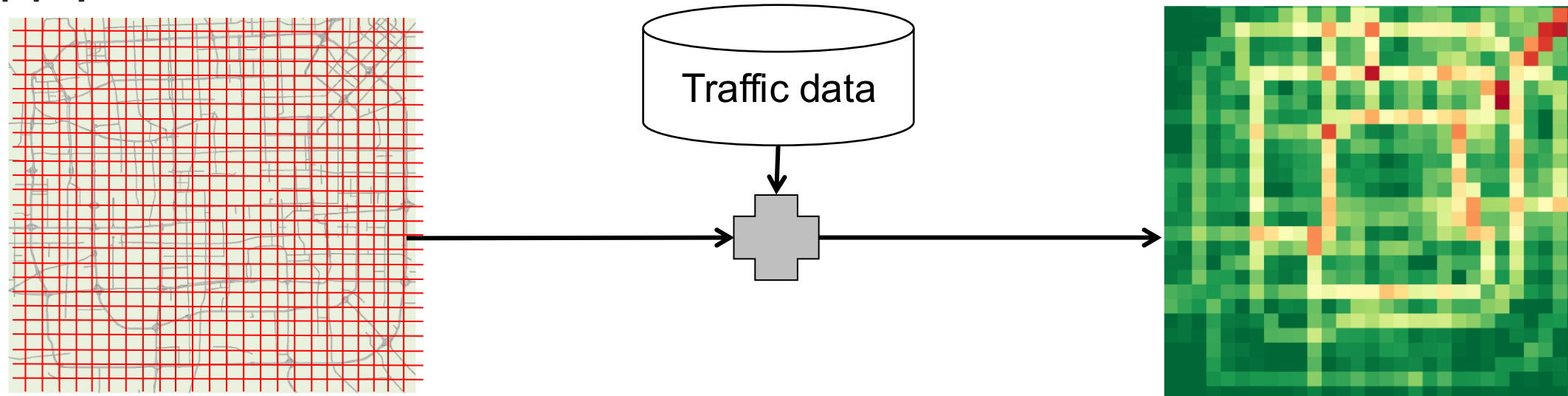
Traffic Prediction

- Traffic prediction algorithms, e.g., ARIMA, takes advantage of repeating occurrences in temporal historical data.
- Deep neural networks are being increasingly used, to capture complex non-linear spatial and temporal properties.



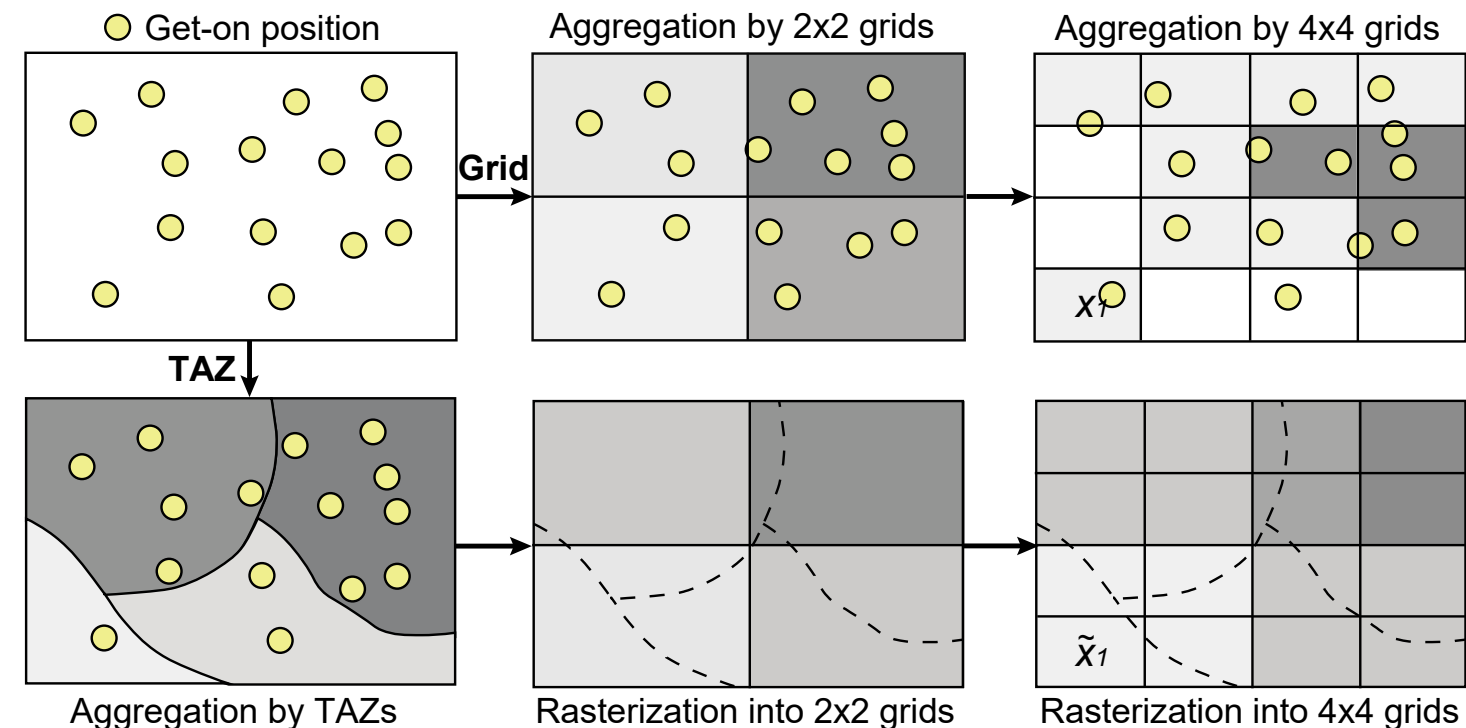
Traffic Prediction

- ST-ResNet [Zhang et al., 2017] predicts in- and out-flow volumes in **regions**: $\{X_t | t \in \{t_0, \dots, t_n\}\} \rightarrow Y_{t_{n+1}}$.
 - partition an underlying territory into grids
 - aggregate in- and out-flows in each grid
 - model the spatio-temporal flows as a sequence of raster images
 - apply convolution-based residual neural network



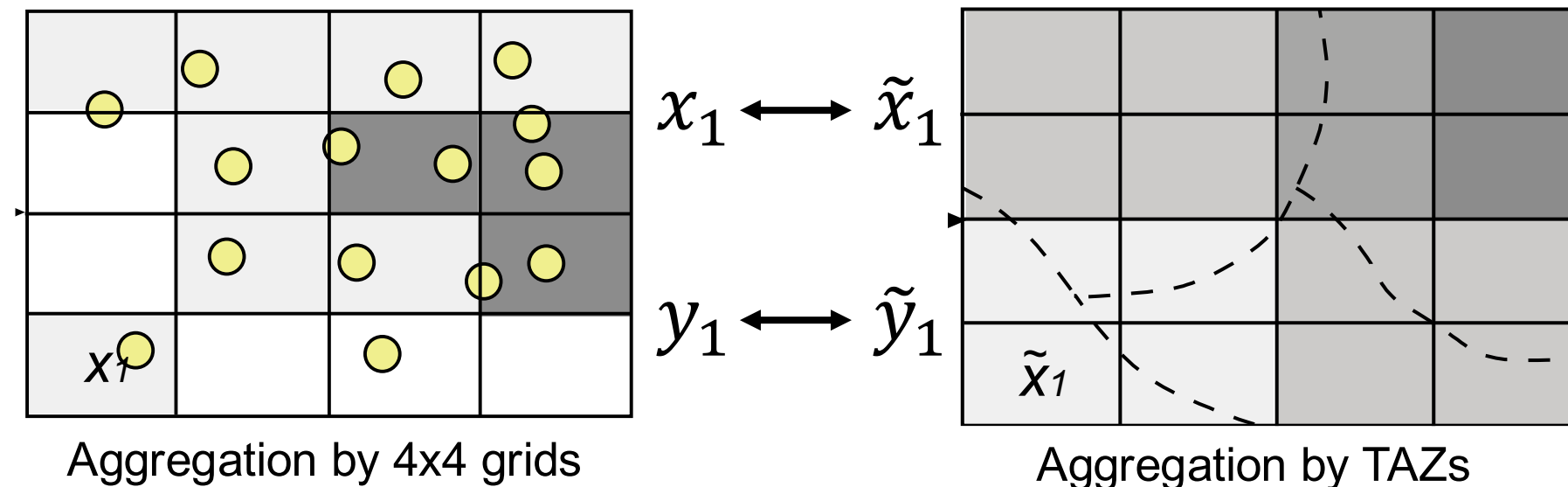
The Modifiable Areal Unit Problem

- Aggregations are influenced by both **partition shapes** and **scales**
 - Shape: grids vs. administrative units
 - Scale: coarse vs. fine granularity
 - Can be regarded as an example of Simpson's paradox
- Adversarial perturbation problem of DNNs.
 - The input difference may be marginal, but it may cause significant effects on the outputs.



Research Goals

- In consultation with a collaborating researcher in geography and traffic analysis, we distill three research goals:
 - G1: Understand the MAUP effect on traffic aggregations
 - G2: Understand the output predictions upon variances in input features
 - G3: Support the exploration of an individual region



Analytical Tasks

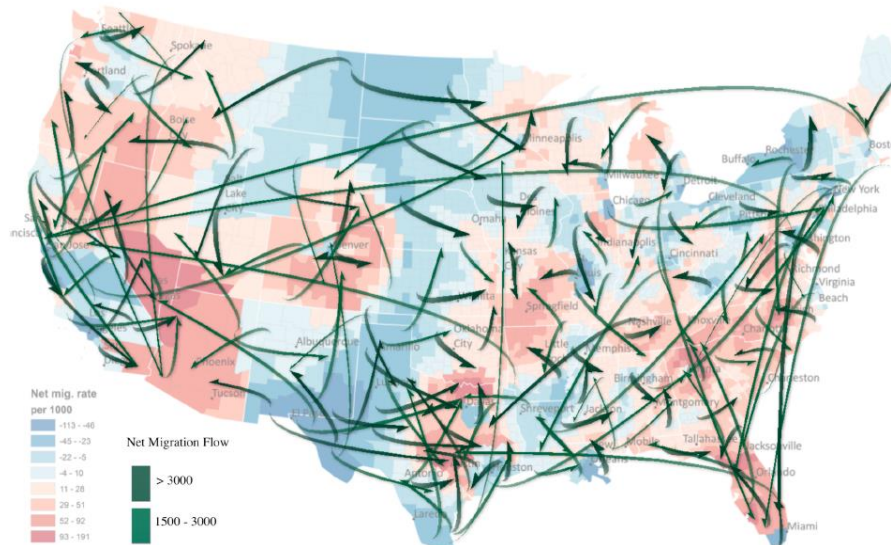
- T1: Spatial variation exploration
 - Traffic distribution over space (G1 & G3)
 - Output predictions over space (G2 & G3)
- T2: Spatial association analysis
 - Spatial dependency
 - Spatial nonstationarity
- T3: Scale-independent comparison
 - Remove the scaling effect of different attribution ranges
 - $\text{MAE}(x_1 = 100, y_1 = 110) = 10$
 - $\text{MAE}(x_2 = 0, y_2 = 10) = 10$
- Internal working mechanism of DNN is out of the scope.

Related Work

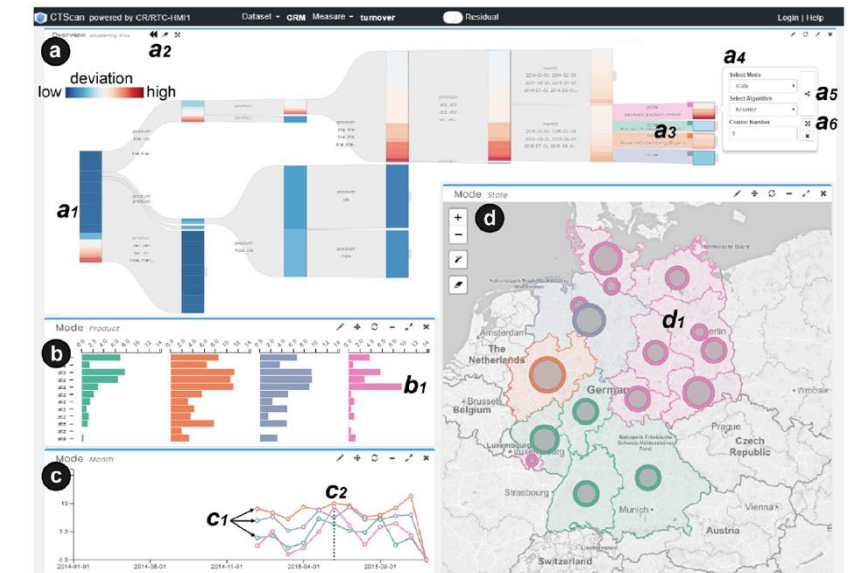
- The MAUP affects data analysis and visualization
 - Spatial generalization and aggregation [Andrienko and Andrienko, 2011]
 - OD flow data smoothing and mapping [Guo and Zhu, 2014]
 - Progressive partition and pattern extraction [Liu et al., 2019]



[Andrienko and Andrienko, 2011]



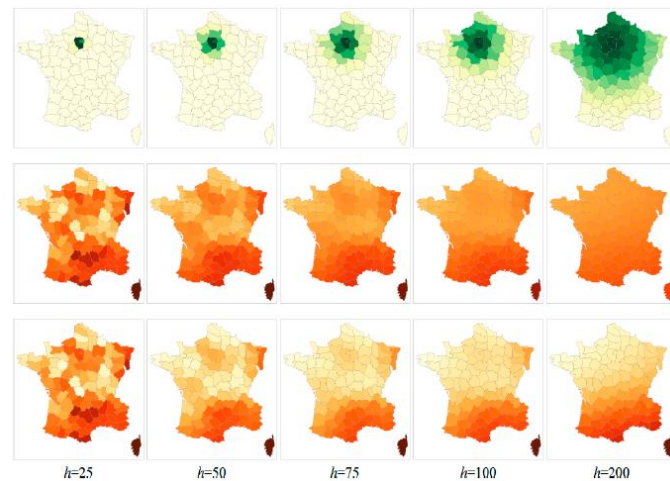
[Guo and Zhu, 2011]



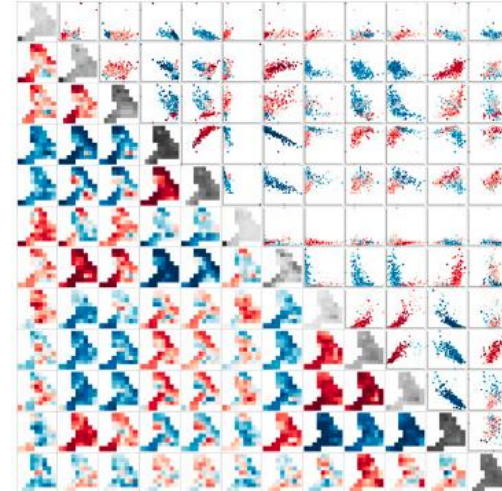
[Liu et al., 2011]

Related Work

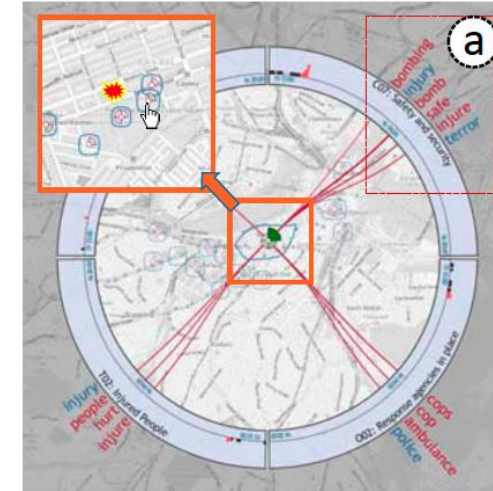
- Many visualizations and visual analytics have been developed to explore and suppress the MAUP.
 - Geographically weighted visualization [Dykes and Brunsdon, 2007]
 - Geo-visual parameter space analysis (geo-vPSA) [Goodwin et al., 2016]
 - Spatial lens [Zhang et al., 2016]
 - Boundary elements [Zhang and Maciejewski, 2017]



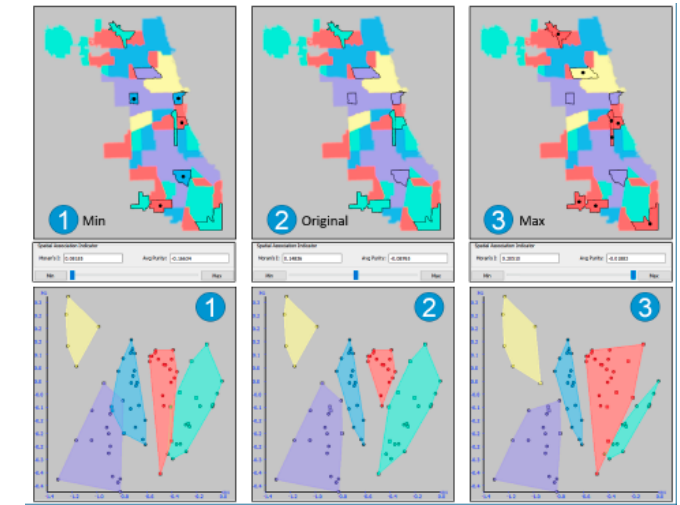
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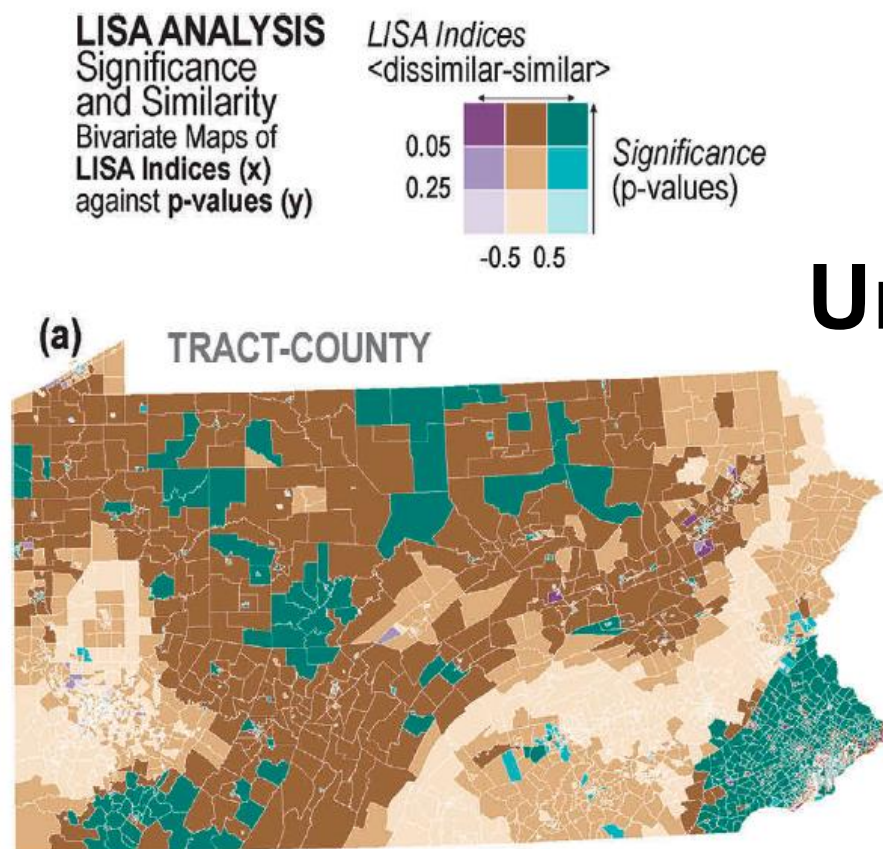
[Zhang and Maciejewski, 2016]

Related Work

- GIS formulates theories for reasoning the MAUP, develop techniques to quantify the MAUP, and identify impacts of the MAUP
 - Moran's I, Geary's C, ...
 - Local indicators of spatial association (LISA)
 - S-maup: Statistical test to measure the sensitivity to the MAUP
- Visualization techniques are on a high demand [Nelson and Brewer, 2017].
- Existing visualizations, however, are limited to ...
 - the newness of the techniques
 - the complexity of displaying such a large and diverse amount of spatially represented data

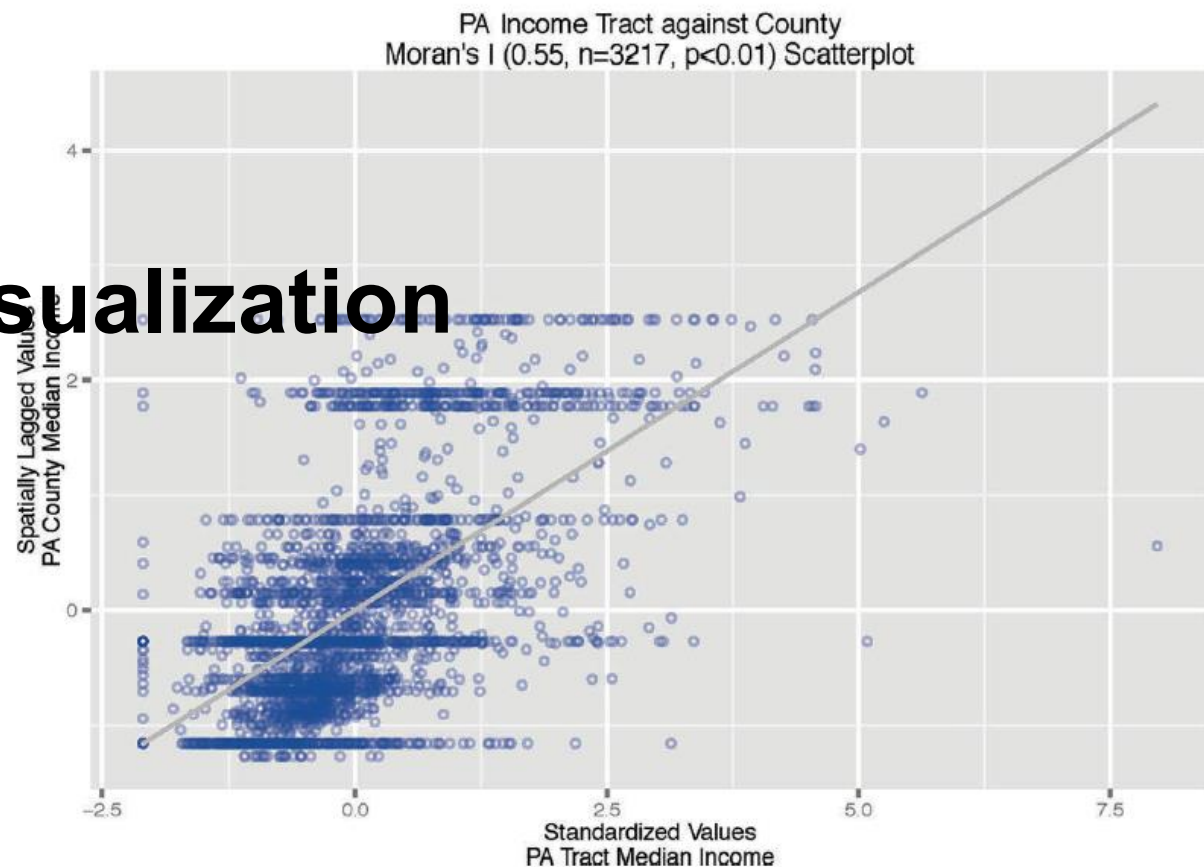
Related Work

- Exploratory spatial data analytical (ESDA) approach to understand the scalar effects of the MAUP [Nelson and Brewer, 2017]



bivariate choropleth map

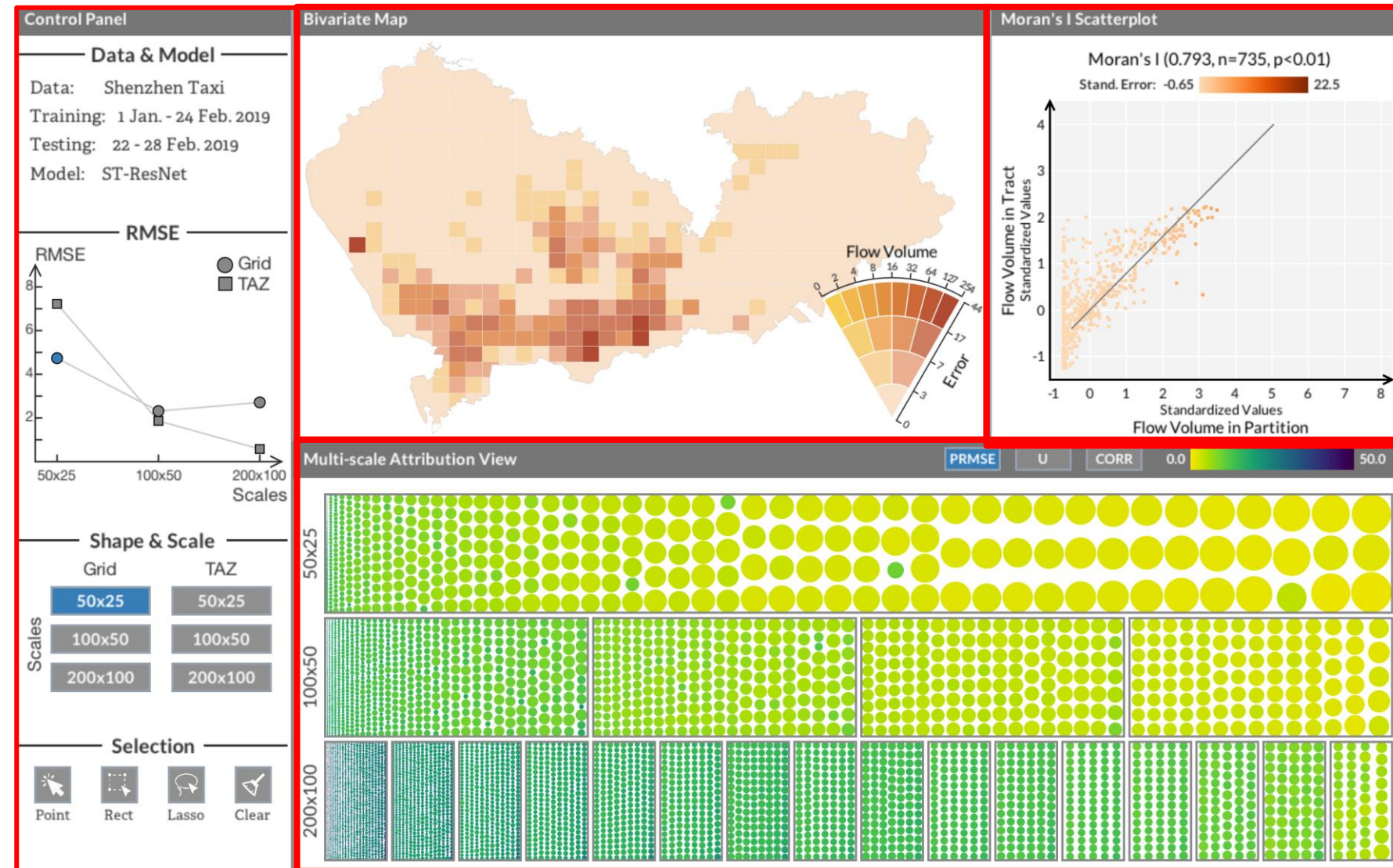
Unit Visualization



Moran's I scatterplot

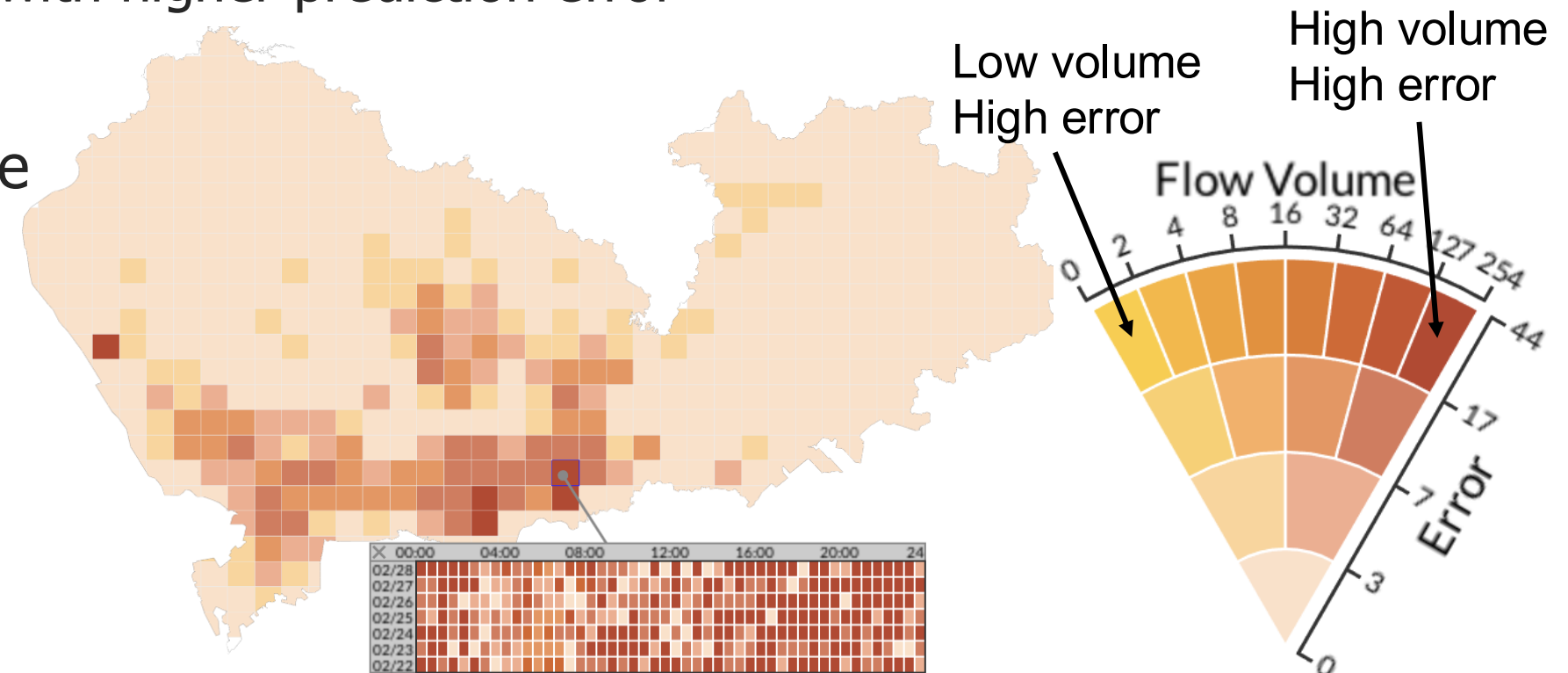
Visualization Design

- Design rationales
 - Coordinated Views
 - Overview + Detail
 - Unit Visualization
- We develop a CMV system
 - Control Panel
 - Bivariate Map
 - Moran's I Scatterplot
 - Multi-scale Attribution View



Bivariate Map

- To support spatial variation exploration (T1)
 - Bivariate map to show traffic volume and prediction error simultaneously
 - Value-Suppressing Uncertainty Palette (VSUP)
 - Emphasizes those grids with higher prediction error
- Temporal View to show temporal variations of all time stamps of a selected region



Moran's I Scatterplot

- To reveal the spatial autocorrelation of traffic volumes in each grid-based partition and local tract (T2)

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} (x_g^i - \bar{x}_g^{ij})(x_g^j - \bar{x}_g^{ij})}{\sum_i (x_g^i - \bar{x}_g^{ij})^2}$$

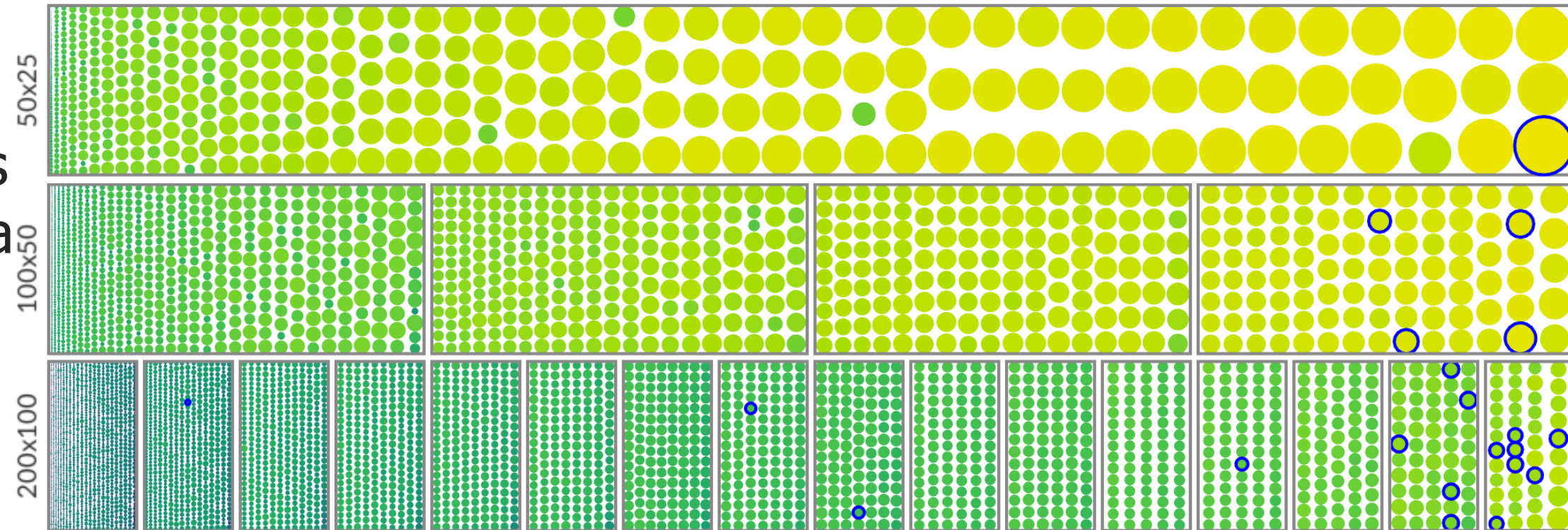
- Queen contiguity spatial weight matrix
- 3 × 3 matrix size – convolution kernel size by ST-ResNet
- Decompose the global Moran's I into local indicators of spatial association (LISA) indices



Multi-scale Attribution View

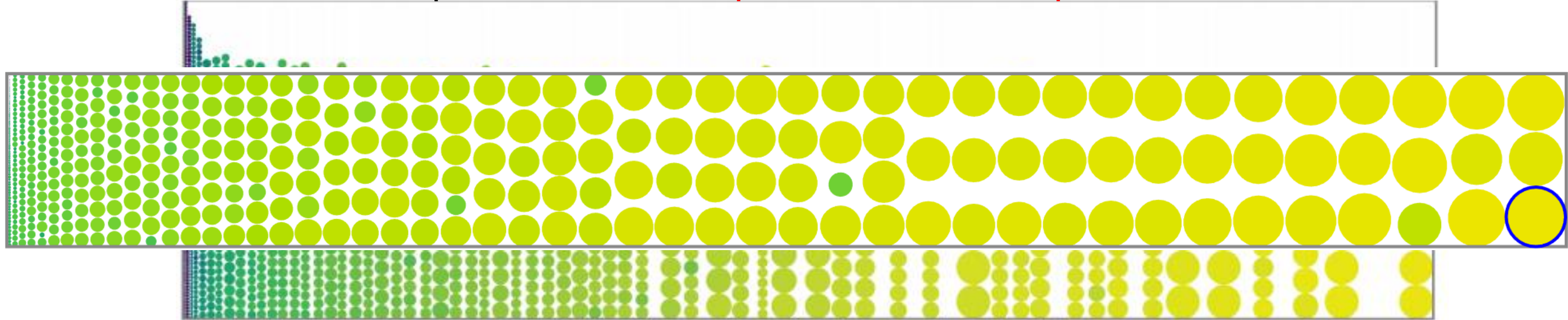
- To support scale-independent comparison (T3).
 - Position: sorted in ascending order by **absolute error**
 - Size: size encodes **traffic volume**
 - Color: color code each dot according **scale-independent metrics**

- Three plots of partition scales in a hierarchical structure to support comparison.



Multi-scale Attribution View

- Alternative design: non-linear dot plots [Rodrigues & Weiskopf, 2018]
 - Position: horizontal position based on prediction error. **Space usage deficiency**
 - Size: adjusted based on the number of dots in each stack. **Misleading correlation analysis**
 - Color: scale-independent metrics. **Spoil multi-scale comparison**

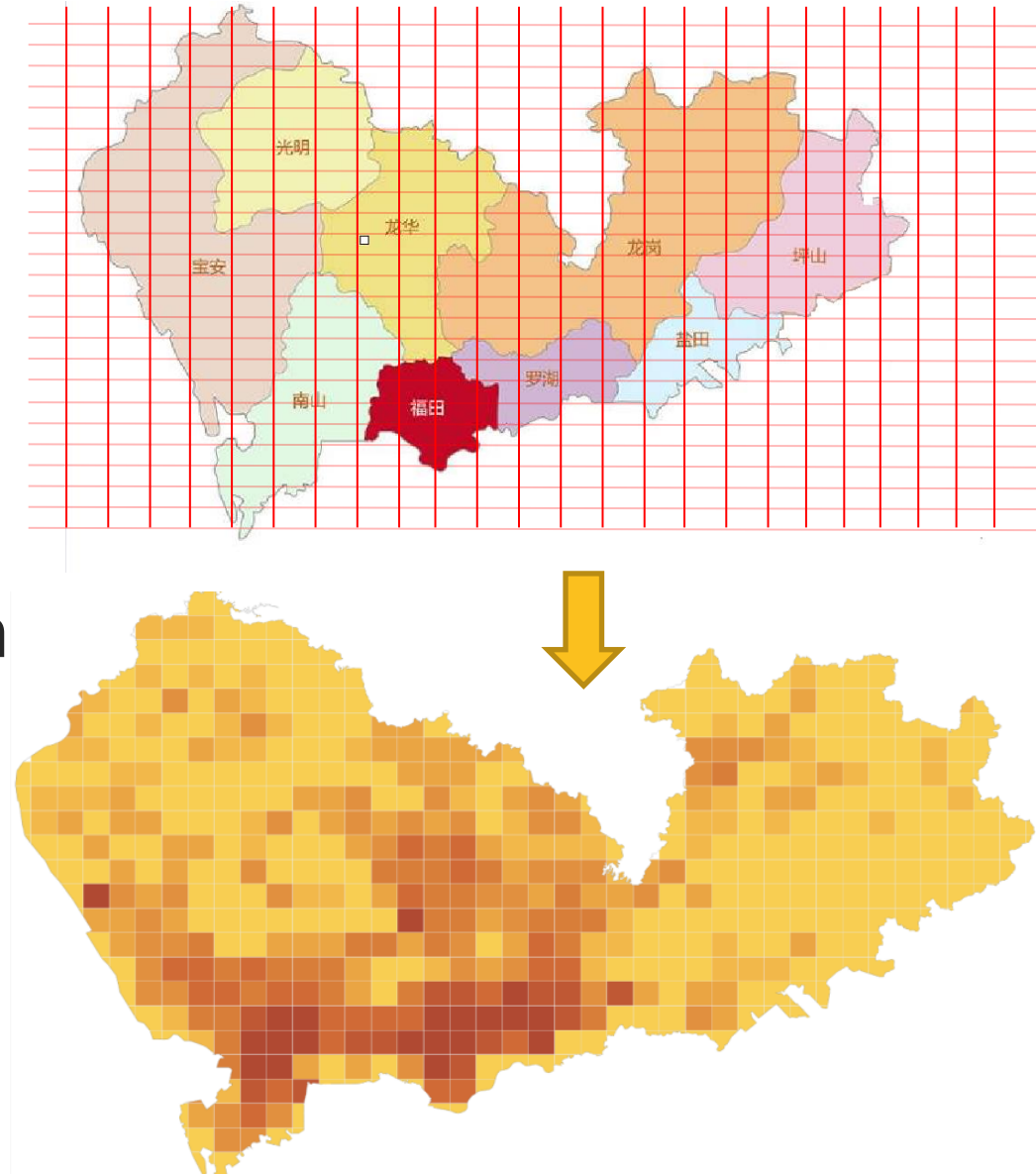


(b) Nonlinear dot plot in constant height

N. Rodrigues and D. Weiskopf. Nonlinear dot plots. IEEE TVCG, 24(1):616–625, 2018.

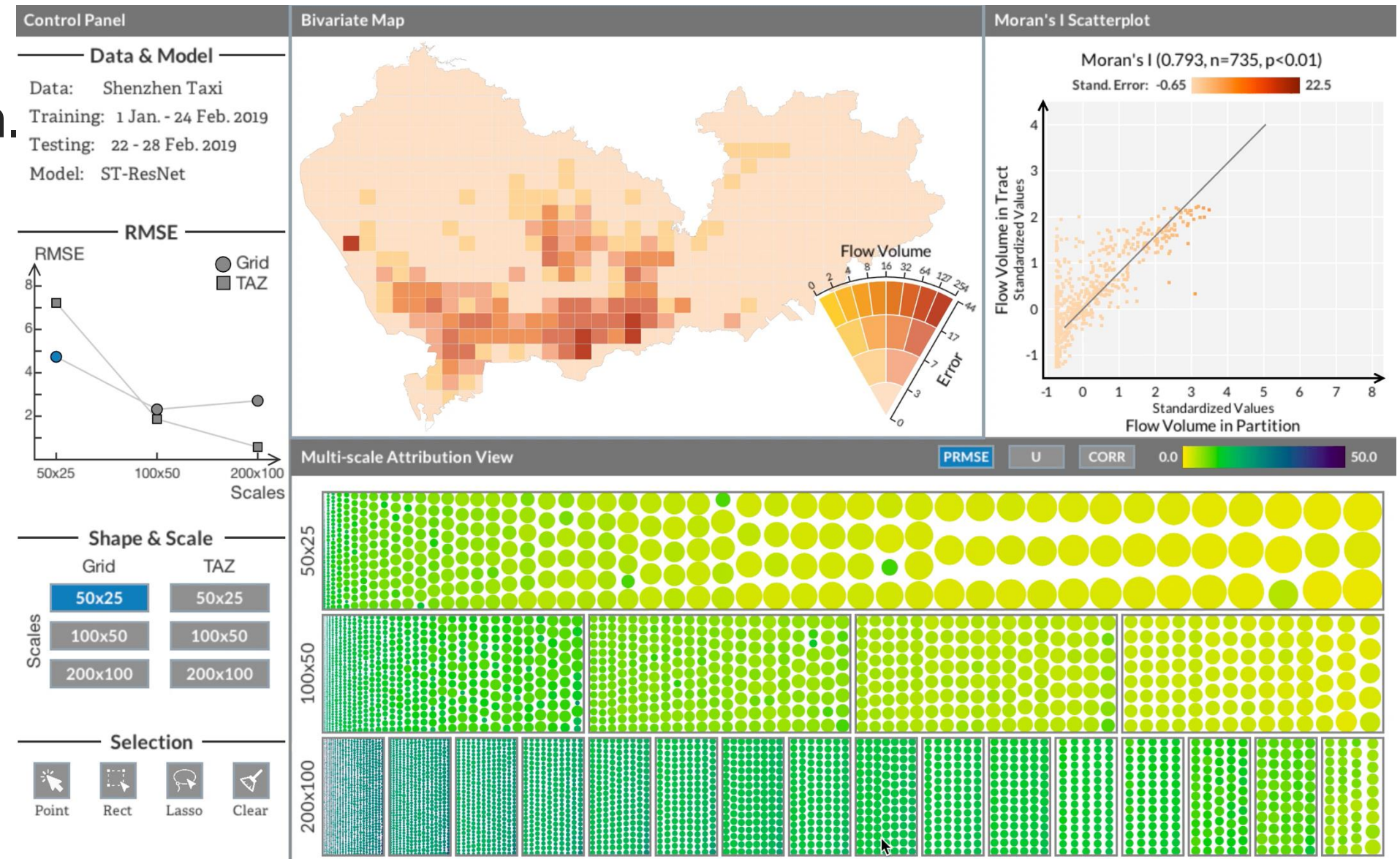
Data and Models

- Dataset
 - Taxi transaction records from 01 Jan. 2019 to 28 Feb. 2019 (59 days)
 - Traffic analysis zones (TAZs): 1066
- Geographical partition
 - Shapes: by grids vs. by TAZs followed by rasterization
 - Scales: 50x25, 100x50, 200x100
- ST-ResNet models
 - 6 models (2 shapes x 3 scales)
 - 52-days data as training, 7-days data as testing



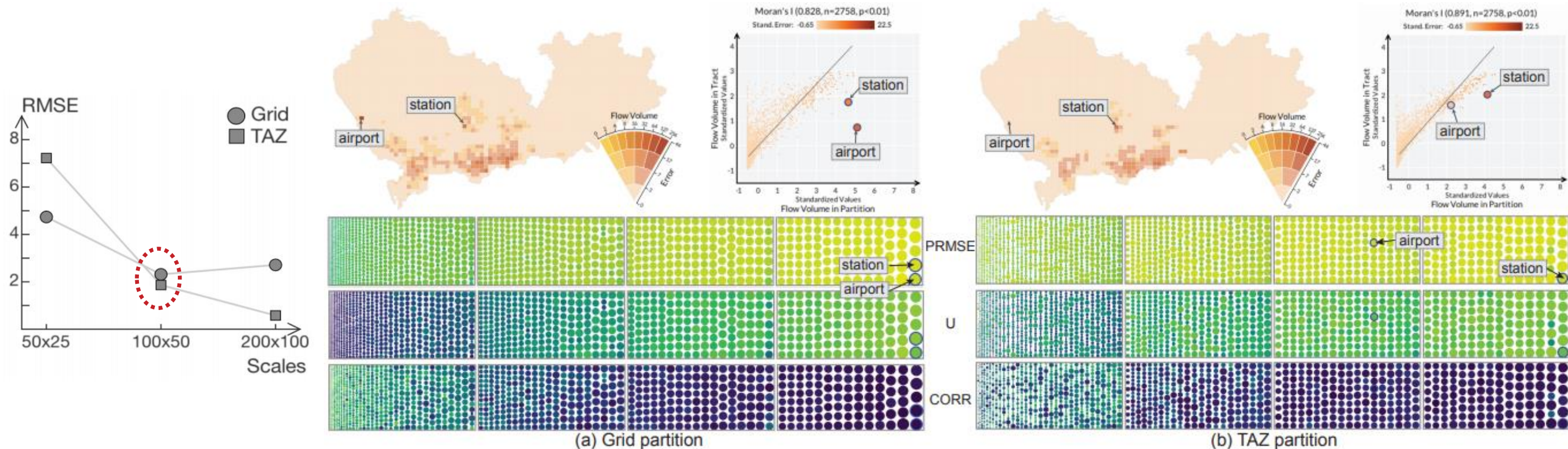
Study 1: Diagnostics of Multi-scale Predictions

- Smallest RMSE at scale 100x50 for grid partition.
- Instead, scale-independent metrics show performance at scale 50x25 is best.
 - Smaller PRMSE
 - Smaller uncertainty coefficient
 - Bigger correlation coefficient



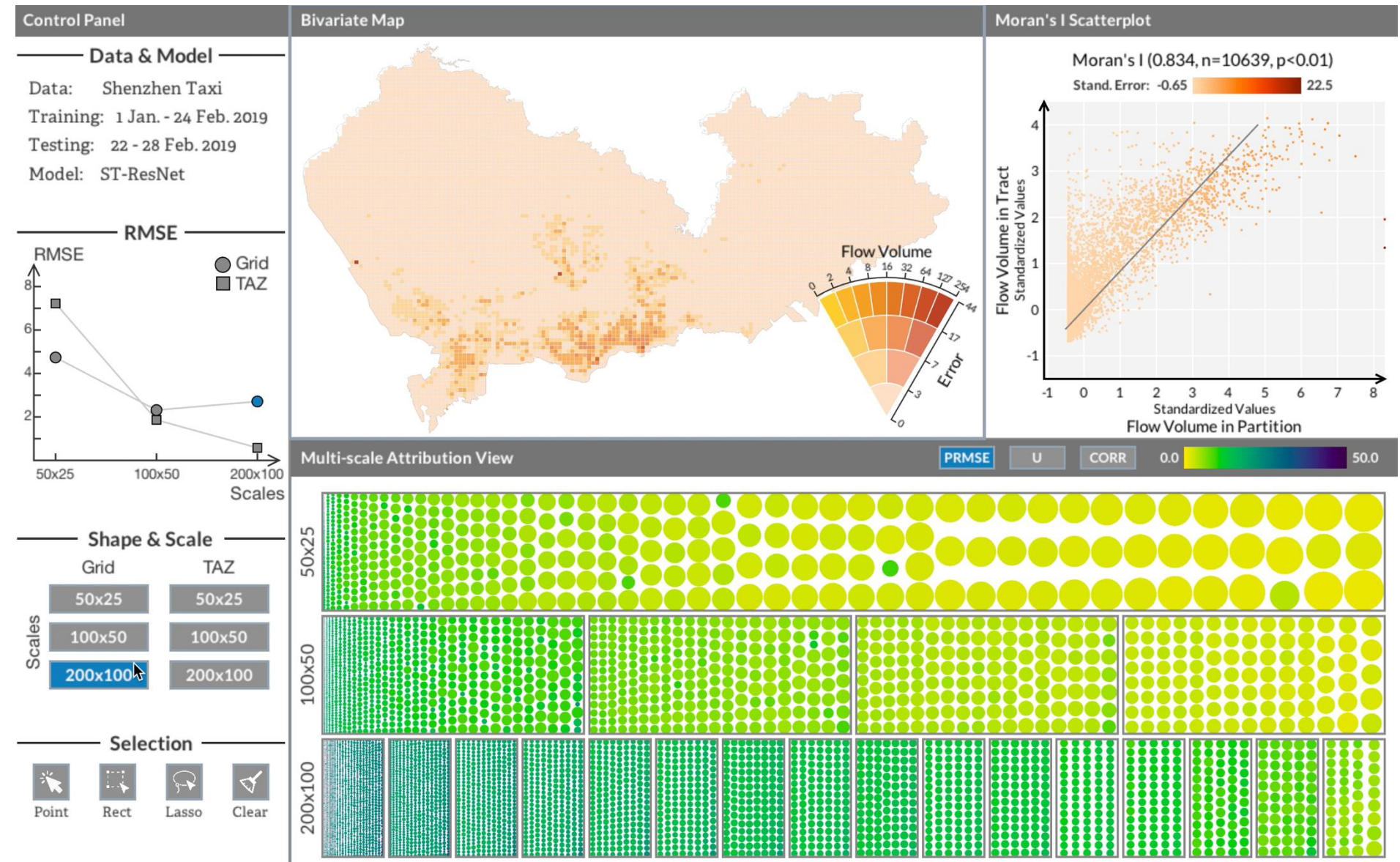
Study 2: Comparison of Different Partition Shapes

- RMSEs are similar for grid and TAZ partitions at scale 100x50.
- Scale-independent metrics show TAZ partition achieves better performance.
 - especially for the airport region



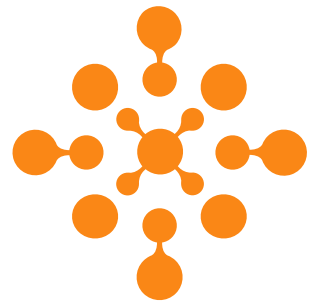
Study 3: Investigation of Individual Units

- With TAZ partition, predictions for **the airport** region are significantly improved, whilst those for **the station** are not improved much.
- TAZ for the airport region is big – traffic volume is smoothed.



Conclusion

- Lessons learned
 - Fine-grained partition scales may generate worse scale-independent metrics.
 - Finer partitions may cause non-convergent DNN models.
 - RMSE may not be a good indicator – scale-independent metrics are more appropriate for multi-scale comparison.
 - TAZ partition is more suitable than Grid partition for deep traffic prediction.
 - Partition methods (e.g., gradual partition) proposed in GIS and transportation can improve network prediction – encourage cross-disciplinary collaboration.
- Future work
 - Graph convolutional networks
 - Attention mechanism



VIS 2020

谢谢！ Thank You!

Paper: <https://arxiv.org/abs/2007.15486>

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